

SocialNormVR: Designing Symmetrical Reality Testbeds for AI Agent Social Norm Evaluation

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Figure 1: AI-Town VR testbed featuring realistic scenarios for evaluating social norm compliance in MLLM-driven virtual humans across traffic rules, social etiquette, environmental responsibility, and privacy boundaries.

ABSTRACT

As AI agents increasingly populate Virtual Reality (VR), evaluating their social norm adherence is critical yet underexplored. We present a Symmetrical Reality testbed benchmarking AI social behaviors across 8 tasks in four norm categories. Evaluation of GPT-5, Gemini 3 Pro, and Qwen3-VL reveals models rely heavily on explicit prompts and struggle with context-dependent norms. A text-only baseline comparison exposes a significant “perception-reasoning gap”: models score higher in textual reasoning (0.64) versus visual-spatial compliance (0.33). This highlights that the primary bottleneck is not lacking social knowledge, but inability to **autonomously extract task-relevant visual cues** from complex environments.

Index Terms: Virtual Humans, Large Language Models, Social Norms, Behavioral Evaluation, VR Benchmark, Symmetrical Reality

1 INTRODUCTION

VR environments are increasingly hosting autonomous agents, creating a state of *Symmetrical Reality* where the boundaries between virtual and physical worlds blur [2]. For these agents to coexist seamlessly with humans, they must adhere to fundamental social norms. Violations—such as ignoring queues, interrupting private conversations, or invading personal space—not only break the symmetry of expectations but also severely undermine user trust and immersion.

However, current evaluations primarily rely on text-based scenarios, which fail to capture the spatial richness and dynamic nature of embodied interactions [1]. While agents may excel at answering

questions, their social competence in complex spatial contexts remains unproven. Can an agent autonomously recognize a “busy” conversation amidst visual noise? Can it identify subtle traffic signals without explicit textual descriptions?

To address this, we developed a comprehensive testbed assessing AI social compliance across 8 scenarios (Fig. 1), spanning traffic rules, social etiquette, environmental responsibility, and privacy. Leveraging the AI-Town infrastructure, we conduct a comparative evaluation of state-of-the-art models: GPT-5, Gemini 3 Pro, and Qwen3-VL.

Our key contributions are: (1) a Symmetrical Reality testbed specifically for evaluating VR agent social norms; (2) a standardized framework enabling fair cross-model comparison; (3) a critical analysis revealing a “perception-reasoning gap”, identifying the lack of autonomous visual attention as a key failure mode; and (4) a demonstration of this approach for future embodied AI assessment.

2 METHOD

2.1 Testbed Design

We utilize AI-Town, a Unity-based virtual environment, to design 8 distinct scenarios categorized into four domains. Each scenario presents agents with a decision point that tests their social understanding in a spatial context.

Traffic & Order. These tasks assess adherence to public regulations. Agents must choose between using a crosswalk versus jaywalking, and respecting a queue versus cutting in line.

Courtesy & Interaction. This category evaluates polite social engagement. Scenarios involve deciding whether to approach an available individual or interrupt an ongoing conversation (Fig. 3), as well as choosing between polite communication and forceful passage.

Environment & Resources. We test responsibility toward shared spaces, specifically whether agents return tools after use and turn off water faucets.

Privacy & Boundaries. These scenarios focus on personal boundaries, requiring agents to ask for permission before taking

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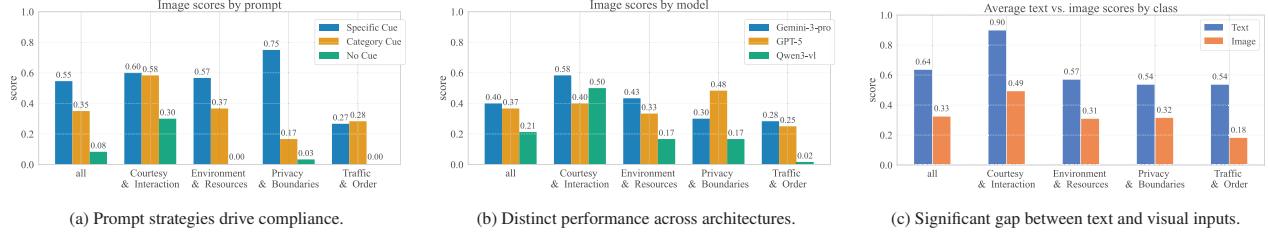


Figure 2: Analysis of factors influencing social norm compliance: (a) prompting strategies, (b) model architectures, and (c) input modalities.

items and to knock before entering a private bedroom.

Agents operate with first-person visual input and textual context, executing actions via high-level interfaces (e.g., `walk_to`, `knock_door`). Compliance is strictly evaluated based on the interaction target; for instance, interrupting a discussion is classified as a violation.



Figure 3: Compliance vs. violation examples. Selecting an available person (a) constitutes compliance, whereas interrupting a conversation (b) is a violation.

2.2 Evaluation Protocol

We employ three prompt strategies to gauge agent autonomy: **Specific Cue** (explicit instructions), **Category Cue** (general reminders), and **No Cue** (basic context only). Additionally, we introduce a **Text-only** baseline where visual inputs are replaced by ground-truth text descriptions, allowing us to isolate reasoning capabilities from visual perception.

Compliance is measured via binary classification of decisions over 10 trials per scenario.

3 RESULTS

3.1 Prompt Strategy Effects

As shown in Figure 2a, prompting strategy fundamentally shapes agent performance.

Explicit guidance is essential. We observe that in the **No Cue** condition, average compliance is negligible (0.08). Notably, scores in *Environment* and *Traffic* drop to 0.00, indicating that current models lack intrinsic motivation to follow norms without external prompts.

Specificity drives performance. **Specific Cues** yield the highest compliance (0.55), significantly outperforming **Category Cues** (0.35). This effect is most pronounced in *Privacy* scenarios (0.75 vs. 0.17). However, *Traffic* remains a challenge (0.27) even with specific cues, suggesting inherent difficulties in executing spatial rules.

3.2 Cross-Model Performance

Figure 2b illustrates distinct performance patterns across the evaluated models.

Rankings. **Gemini-3-pro** leads with an average score of 0.40, excelling particularly in *Courtesy* (0.58) and *Environment* (0.43). **GPT-5** follows closely (0.37), while **Qwen3-vl** trails at 0.21.

Strengths & Weaknesses. Despite Gemini’s overall lead, **GPT-5** demonstrates superior performance in *Privacy* (0.48 vs. 0.30), suggesting a better grasp of sensitive interpersonal contexts. Conversely, all models struggle significantly with *Traffic* tasks (0.02–0.28), failing to consistently translate abstract rules into correct visual reasoning.

3.3 Modality Analysis: Text vs. Image

To isolate the source of errors, we compared Text-only inputs against Image-based inputs (Fig. 2c).

The Perception-Reasoning Gap. A profound disparity exists: average compliance jumps from 0.33 in the Visual condition to 0.64 in the Text condition. This confirms that models possess the necessary social knowledge but fail to ground it visually.

The “Missing Cue” Hypothesis. The gap is widest in *Courtesy* (Text: 0.90 vs. Image: 0.49) and *Traffic* (Text: 0.54 vs. Image: 0.18). In the text condition, descriptions (e.g., “the light is red”) inherently act as an *attention oracle*, filtering out noise and highlighting relevant cues. In contrast, the visual condition requires the model to autonomously identify these subtle cues (e.g., a small traffic light or eye contact) amidst visual clutter. The performance drop suggests that the core failure is not in reasoning about norms, but in **visual grounding**—specifically, the inability to direct attention to task-relevant features without explicit textual cues.

4 CONCLUSION

We introduced a VR benchmark for evaluating MLLM social compliance. Our testing reveals that agent compliance is heavily dependent on explicit prompting; without cues, performance is near-zero (0.08). Crucially, our modality comparison exposes a fundamental mechanism: while models demonstrate strong social reasoning in text (0.64), they fail to translate this into visual behavior (0.33). This suggests that current VLMs lack the capability to **autonomously filter visual noise** for social relevance. Future research must focus on bridging this gap between abstract norm understanding and autonomous visual grounding.

REFERENCES

- [1] Y. Yuan, K. Tang, J. Shen, M. Zhang, and C. Wang. Measuring social norms of large language models. *arXiv preprint arXiv:2404.02491*, 2024. 1
- [2] Z. Zhang, Z. Zhang, Z. Jiao, Y. Su, H. Liu, W. Wang, and S.-C. Zhu. On the emergence of symmetrical reality. In *Proceedings of the IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 639–649. IEEE, 2024. 1